

Create Database

The screenshot displays the MongoDB Cloud Portal interface. The left sidebar shows the 'Data Explorer' with a tree view of clusters and databases. The main area displays the 'customers' collection in the 'Bankdb' database. The table shows 8 documents with columns: _id, customer_id, name, city, account_type, and balance. The bottom status bar indicates the alert system is 'All Good' and shows the date '01-09-2026'.

	_id	customer_id	name	city	account_type	balance
1	ObjectId('6a969cda9427a4...')	101	"Ravi Kumar"	"Hyderabad"	"Savings"	85000
2	ObjectId('6a969ceb9427a4...')	102	"Priya Sharma"	"Vijayawada"	"Current"	125000
3	ObjectId('6a969cfc9427a4...')	103	"Arjun Reddy"	"Hyderabad"	"Savings"	45000
4	ObjectId('6a96a0299427a4...')	104	"Sneha Rao"	"Warangal"	"Savings"	210000
5	ObjectId('6a96a03c9427a4...')	105	"Kiran Kumar"	"Guntur"	"Current"	65000
6	ObjectId('6a96a0499427a4...')	106	"Anjali Singh"	"Hyderabad"	"Savings"	175000
7	ObjectId('6a96a05a9427a4...')	107	"Rahul Verma"	"Bengaluru"	"Current"	95000
8	ObjectId('6a96a0649427a4...')	108	"Meena Devi"	"Chennai"	"Savings"	320000

PART 3: INSERT CUSTOMER DATA

Cluster0 > bank_DB > customers

View monitoring Visualize Your Data

Documents 8 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

25 1 - 8 of 8

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	Edit Duplicate Delete
2	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"	Edit Duplicate Delete
3	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	Edit Duplicate Delete
4	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	Edit Duplicate Delete
5	ObjectId('6a8965bfc4c890d...	105	"Kiran Kumar"	"Guntur"	Edit Duplicate Delete
6	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	Edit Duplicate Delete
7	ObjectId('6a8965bfc4c890d...	107	"Rahul Verma"	"Bengaluru"	Edit Duplicate Delete
8	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	Edit Duplicate Delete

PART 4: INSERT TRANSACTION DATA

Cluster0

customers

transactions

+

Cluster0 > bank_DB > transactions

View monitoring

Visualize Your Data

Documents 15

Aggregations

Schema

Indexes 1

Validation

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Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options ▶

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25 ▼

1 - 15 of 15

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{ }

🏠

🏠 transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	
1	ObjectId('6a8966e0c4c890d...	1001	101	"Deposit"	✎ 📄 🗑️
2	ObjectId('6a8966e0c4c890d...	1002	101	"Withdrawal"	✎ 📄 🗑️
3	ObjectId('6a8966e0c4c890d...	1003	102	"Deposit"	✎ 📄 🗑️
4	ObjectId('6a8966e0c4c890d...	1004	102	"Withdrawal"	✎ 📄 🗑️
5	ObjectId('6a8966e0c4c890d...	1005	103	"Deposit"	✎ 📄 🗑️
6	ObjectId('6a8966e0c4c890d...	1006	103	"Withdrawal"	✎ 📄 🗑️
7	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"	✎ 📄 🗑️
8	ObjectId('6a8966e0c4c890d...	1008	104	"Withdrawal"	✎ 📄 🗑️

QUERY 2: FIND ACTIVE CUSTOMERS

Cluster0

customers

transactions

+

Cluster0 > bank_DB > customers

View monitoring

Visualize Your Data

Documents 8

Aggregations

Schema

Indexes 1

Validation

🕒

{ city: "Hyderabad" }

Generate query ↗

Explain

Reset

Find

Options ▶

➕ ADD DATA

📄 BULK

📄 EXPORT CODE

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25 1 - 3 of 3 ↺ ⏪ ⏩ ⏴ ⏵ ⏶ ⏷ ⏸ ⏹

🏠 customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	✎ 📄 🗑
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	✎ 📄 🗑
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	✎ 📄 🗑

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⏴ ⏵ ⏶ ⏷ ⏸ ⏹

QUERY 3: FIND CUSTOMERS WITH BALANCE GREATER THAN 100000

PROJECT
Project 0

Cluster0

Cluster0 > bank_DB > customers

Documents 8 Aggregations Schema Indexes 1 Validation

`{
 "balance": {
 "$gt": 100000
 }
}`

Generate query Explain Reset Find Options

ADD DATA BULK EXPORT CODE

25 1 - 4 of 4

customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"	
2	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	
4	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	

QUERY 4: FIND HYDERABAD CUSTOMERS

Cluster0 customers transactions +

Cluster0 > bank_DB > customers [View monitoring](#) [Visualize Your Data](#)

Documents 8 Aggregations Schema Indexes 1 Validation

{ "city": "Hyderabad" } [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[+ ADD DATA](#) [BULK](#) [EXPORT CODE](#) 25 1 - 3 of 3 [Refresh](#) [Previous](#) [Next](#) [Menu](#) [Code](#) [Grid](#)

customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	Edit Copy Share Delete
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	Edit Copy Share Delete
3	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	Edit Copy Share Delete

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QUERY 5: FIND SAVINGS ACCOUNT CUSTOMERS

Filter:

```
{
  "account_type": "Savings"
}
```

Cluster0 > bank_DB > customers [View monitoring](#) [Visualize Your Data](#)

[Documents](#) 8 [Aggregations](#) [Schema](#) [Indexes](#) 1 [Validation](#)

{ "account_type": "Savings" } [Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[+ ADD DATA](#) [BULK](#) [EXPORT CODE](#) 25 1 - 5 of 5 [Refresh](#) [Previous](#) [Next](#) [Menu](#) [JSON](#) [Table](#)

customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	Edit Copy Share Delete
2	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	Edit Copy Share Delete
3	ObjectId('6a8965bfc4c890d...	104	"Sneha Rao"	"Warangal"	Edit Copy Share Delete
4	ObjectId('6a8965bfc4c890d...	106	"Anjali Singh"	"Hyderabad"	Edit Copy Share Delete
5	ObjectId('6a8965bfc4c890d...	108	"Meena Devi"	"Chennai"	Edit Copy Share Delete

QUERY 6: FIND TRANSACTIONS GREATER THAN 20000

Open:
transactions

Cluster0 > bank_DB > transactions

View monitoring

Visualize Your Data

Documents 15

Aggregations

Schema

Indexes 1

Validation

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{
 "amount": {
 "\$gt": 20000
 }
}

Generate query ↗

Explain

Reset

Find

Options ▶

➕ ADD DATA ▾

📄 BULK ▾

📄 EXPORT CODE

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25 ▾ 1 - 9 of 9 ↺ ⏪ ⏩ ▾ ⌵ ⌴ ⌶

🏠 transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	
1	ObjectId('6a8966e0c4c890d...	1001	101	"Deposit"	✎ 📄 🗑
2	ObjectId('6a8966e0c4c890d...	1003	102	"Deposit"	✎ 📄 🗑
3	ObjectId('6a8966e0c4c890d...	1005	103	"Deposit"	✎ 📄 🗑
4	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"	✎ 📄 🗑
5	ObjectId('6a8966e0c4c890d...	1008	104	"Withdrawal"	✎ 📄 🗑

Filter:

Filter:

Filter:

Cluster0
customers
transactions
+

Cluster0 > bank_DB > transactions
View monitoring
Visualize Your Data

Documents 15
Aggregations
Schema
Indexes 1
Validation

{
"mode": "ATM"
}
Generate query
Explain
Reset
Find
Options

ADD DATA
BULK
EXPORT CODE
25
1 - 7 of 7

	_id	ObjectId	transaction_id	Int32	customer_id	Int32	mode	String
1		ObjectId('6a8966e0c4c890d...	1002		101		"ATM"	
2		ObjectId('6a8966e0c4c890d...	1004		102		"ATM"	
3		ObjectId('6a8966e0c4c890d...	1006		103		"ATM"	
4		ObjectId('6a8966e0c4c890d...	1008		104		"ATM"	
5		ObjectId('6a8966e0c4c890d...	1011		106		"ATM"	
6		ObjectId('6a8966e0c4c890d...	1013		107		"ATM"	

=====

QUERY 9: FIND DEPOSITS GREATER THAN 50000

=====

Filter:

Cluster0 > bank_DB > transactions

View monitoring

Visualize Your Data

Documents 15

Aggregations

Schema

Indexes 1

Validation

```
{
  "type": "Deposit",
  "amount": {
    "$gt": 50000
  }
}
```

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 3 of 3

transactions

	_id ObjectId	transaction_id Int32	customer_id Int32	type String	
1	ObjectId('6a8966e0c4c890d...	1007	104	"Deposit"	
2	ObjectId('6a8966e0c4c890d...	1010	106	"Deposit"	
3	ObjectId('6a8966e0c4c890d...	1014	108	"Deposit"	

QUERY 10: FIND HYDERABAD CUSTOMERS WITH HIGH BALANCE

Cluster0 customers transactions +

Cluster0 > bank_DB > customers [View monitoring](#) [Visualize Your Data](#)

Documents 8 Aggregations Schema Indexes 1 Validation

```
{
  "city": "Hyderabad",
  "balance": {
    "$gt": 100000
  }
}
```

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

[ADD DATA](#) [BULK](#) [EXPORT CODE](#) 25 1-1 of 1

customers

	customer_id Int32	name String	city String	account_type String	
1	106	"Anjali Singh"	"Hyderabad"	"Savings"	edit share copy delete

QUERY 11: FIND CUSTOMERS IN MULTIPLE CITIES

Filter:

Cluster0

customers

transactions

+

Cluster0 > bank_DB > customers

View monitoring

Visualize Your Data

Documents 8

Aggregations

Schema

Indexes 1

Validation

```
{
  "city": {
    "$in": [
      "Hyderabad",
      "Vijayawada",
      "Warangal"
    ]
  }
}
```

Generate query

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

25

1 - 5 of 5

customers

	_id ObjectId	customer_id Int32	name String	city String	
1	ObjectId('6a8964e3c4c890d...	101	"Ravi Kumar"	"Hyderabad"	
2	ObjectId('6a8965bfc4c890d...	102	"Priya Sharma"	"Vijayawada"	
3	ObjectId('6a8965bfc4c890d...	103	"Arjun Reddy"	"Hyderabad"	

QUERY 12: SORT CUSTOMERS BY BALANCE

The screenshot shows the MongoDB Atlas Aggregations editor interface. At the top, the breadcrumb navigation indicates the path: Cluster0 > bank_DB > customers. There are buttons for 'View monitoring' and 'Visualize Your Data'. Below the breadcrumb, tabs for 'Documents' (8), 'Aggregations', 'Schema', 'Indexes' (1), and 'Validation' are visible. The 'Aggregations' tab is active, showing a single stage named '\$sort'. The stage configuration is set to 'Stage 1' with a dropdown menu showing '\$sort' and a toggle switch. The left pane displays a JSON document snippet:

```
1 {
2   balance: -1
3 }
```

. The right pane shows the 'Output preview after \$sort stage (Sample of 8 documents)'. It contains two JSON documents. The first document is:

```
{
  "_id": ObjectId('6a8965bfc4c890d41f9f6bce'),
  "customer_id": 108,
  "name": "Meena Devi",
  "city": "Chennai",
  "account_type": "Savings",
  "balance": 320000,
  "status": "Active"
}
```

. The second document is partially visible:

```
{
  "_id": Obj
  "customer
  "name": "
  "city": "
  "account_
  "balance":
  "status":
```

. The interface includes a top bar with 'Generate aggregation', 'Explain', 'Run', and 'Options' buttons. Below the top bar, there are buttons for 'SAVE', 'CREATE NEW', 'EXPORT CODE', 'PREVIEW', 'STAGES', 'TEXT', and 'WIZARD'.

QUERY 13: CALCULATE TOTAL TRANSACTION AMOUNT

Cluster0 > bank_DB > customers View monitoring Visualize Your Data

Documents 8 **Aggregations** Schema Indexes 1 Validation

\$group Generate aggregation Explain Run Options

Untitled - modified SAVE + CREATE NEW EXPORT CODE PREVIEW STAGES TEXT WIZARD

Stage 1 \$group Output preview after \$group stage (Sample of 1 document)

```
1 {
2   "_id": null,
3   "total_amount": {
4     "$sum": "$balance"
5   }
6 }
```

```
_id: null
total_amount: 1120000
```

QUERY 14: CALCULATE AVERAGE TRANSACTION AMOUNT

The screenshot shows the MongoDB Atlas Aggregations editor interface. At the top, the breadcrumb navigation indicates the path: Cluster0 > bank_DB > customers. There are buttons for 'View monitoring' and 'Visualize Your Data'. Below the breadcrumb, tabs for 'Documents' (8), 'Aggregations' (selected), 'Schema', 'Indexes' (1), and 'Validation' are visible. The main workspace has a toolbar with a folder icon, a '\$group' stage selector, a 'Generate aggregation' button with a plus icon, and buttons for 'Explain', 'Run', and 'Options'. Below the toolbar, there's a status bar with 'Untitled - modified', 'SAVE', 'CREATE NEW', 'EXPORT CODE', 'PREVIEW' (checked), 'STAGES', 'TEXT', 'WIZARD', and a settings gear icon. The main area is divided into two panes. The left pane shows the aggregation pipeline for 'Stage 1' with a '\$group' stage. The pipeline is as follows:

```
1 {  
2   "_id": null,  
3   "average_amount": {  
4     "$avg": "$balance"  
5   }  
6 }  
7 |
```

The right pane, titled 'Output preview after \$group stage (Sample of 1 document)', shows the resulting JSON document:

```
_id: null  
average_amount : 140000
```

QUERY 15: TOTAL TRANSACTION AMOUNT BY TYPE

transactions +

Cluster0 > bank_DB > transactions [View monitoring](#) [Visualize Your Data](#)

Documents 15 **Aggregations** Schema Indexes 1 Validation

📁 \$group Edit Explain Run Options ▶

ALL RESULTS `</>` Showing 1 – 2 count results < > ⌵ ⋮ { }

```
{
  "_id": "Deposit",
  "total_amount": 485000
}
```

```
{
  "_id": "Withdrawal",
  "total_amount": 117000
}
```

QUERY 16: NUMBER OF TRANSACTIONS PER CUSTOMER

Untitled - modified   

 PREVIEW    

Stage 1  \$group 

1  {  

2 "_id": "\$customer_id",

3 "transaction_count": {

4 "\$sum": 1

5 }

6 }

Output preview after \$group stage (Sample of 8 documents)

```
_id: 102
transaction_count : 2
```

```
{
  "$sort": {
    "transaction_count": -1
  }
}
```

ALL RESULTS 

Showing 1 - 15 count results     

```
_id: ObjectId('6a8966e0c4c890d41f9f6bd5')
transaction_id: 1005
customer_id: 103
type: "Deposit"
amount: 30000
mode: "UPI"
date: "2026-08-03"
status: "Success"
```

```
_id: ObjectId('6a8966e0c4c890d41f9f6bd2')
transaction_id: 1002
customer_id: 101
type: "Withdrawal"
amount: 10000
mode: "ATM"
```

QUERY 17: CUSTOMER + TRANSACTION JOIN

The screenshot shows the MongoDB Atlas interface for an aggregation pipeline. The pipeline is named "Stage 1: \$lookup" and is currently "ENABLED". The stage is configured with a "lookup" type. The "from" field is set to "customers", and the "localField" is "customer_id". The "foreignField" is "customer_id", and the "as" field is "customer_details". The "pipeline" field is empty. The "stage output" shows a sample of 10 documents, which are the result of the join operation. The documents are structured as follows:

```
1 {
2   "from": "customers",
3   "localField": "customer_id",
4   "foreignField": "customer_id",
5   "as": "customer_details"
6 }
7
```

The "stage output" also shows a sample of 10 documents, which are the result of the join operation. The documents are structured as follows:

```
{
  "_id": ObjectId('6a8966...'),
  "transaction_id": 1005,
  "customer_id": 103,
  "type": "Deposit",
  "amount": 30000,
  "mode": "UPI",
  "date": "2026-08-03",
  "status": "Success",
  "customer_details": {
    "_id": ObjectId('6a8966...'),
    "transaction_id": 1002,
    "customer_id": 101,
    "type": "Withdrawal",
    "amount": 10000,
    "mode": "ATM",
    "date": "2026-08-02",
    "status": "Success",
    "customer_details": {
      "_id": ObjectId('6a8966...'),
      "transaction_id": 1003,
      "customer_id": 102,
      "type": "Deposit",
      "amount": 50000,
      "mode": "NEFT",
      "date": "2026-08-02",
      "status": "Success"
    }
  }
}
```


QUERY 18: DISPLAY CUSTOMER NAME AND TRANSACTION AMOUNT

Cluster0 > bank_DB > transactions

Documents 15 Aggregations Schema Indexes 1 Validation

View monitoring Visualize Your Data

\$lookup \$unwind \$project Explain Run Options ▶

Untitled - modified + CREATE NEW </> PREVIEW {} </> *

Stage 3 \$project

```
1 {
2   "_id": 0,
3   "transaction_id": 1,
4   "customer_name": "$customer_details.name",
5   "city": "$customer_details.city",
6   "type": 1,
7   "amount": 1,
8   "mode": 1
9 }
```

Output preview after \$project stage (Sample of 10 documents)

```
transaction_id : 1005
type : "Deposit"
amount : 30000
mode : "UPI"
customer_name : "Arjun Reddy"
city : "Hyderabad"
```

QUERY 19: TOTAL TRANSACTION AMOUNT FOR EACH CUSTOMER

The screenshot displays a data pipeline configuration interface. At the top, a dropdown menu shows 'Stage 5: \$sort' selected, with an 'ENABLED' toggle and an 'ADD STAGE' button. Below this, a list of stages is visible: Stage 1: \$group, Stage 2: \$lookup, Stage 3: \$unwind, Stage 4: \$project, and Stage 5: \$sort (selected). The 'OPTIONS' dropdown for Stage 5 is open, showing a list of documents. The main area shows the configuration for Stage 5: \$sort, with a dropdown set to '\$sort' and a link to 'Open docs'. The 'STAGE OUTPUT' section shows a sample of 8 documents, each containing 'total_transaction', 'customer_id', and 'customer_name' fields. The documents are sorted by 'total_transaction' in descending order.

Stage 5: \$sort

ENABLED

ADD STAGE

STAGE OUTPUT

Sample of 8 documents

```
1 {
2   "total_transaction": -1
3 }
```

total_transaction	customer_id	customer_name
180	108	"Meena Devi"
120	104	"Sneha Rao"
95	106	"Anjali Singh"
18	108	"Meena Devi"
65	102	"Priya Sharma"
65	102	"Priya Sharma"
52	107	"Rahul Verma"
52	107	"Rahul Verma"

=====

QUERY 20: COMPLEX BANK TRANSACTION REPORT

=====

Stage 5: \$sort

ENABLED

ADD STAGE

STAGE 1: \$match

STAGE 2: \$lookup

STAGE 3: \$unwind

STAGE 4: \$group

STAGE 5: \$sort

OPTIONS

\$sort

Open docs

1 {
2 "total_transactions": -1
3 }

STAGE OUTPUT

Sample of 6 documents

OPTIONS

_id: 108
customer_name: "Meena Devi"
city: "Chennai"
balance: 320000
total_transactions: 1
transaction_count: 2

_id: 104
customer_name: "Sneha Rao"
city: "Warangal"
balance: 210000
total_transactions: 1
transaction_count: 2

_id: 106
customer_name: "Anjali Singh"
city: "Hyderabad"
balance: 175000
total_transactions: 9
transaction_count: 2

_id: 102